Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/20/2022
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	Characterizing Plant Gravity Perception Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	Delaware	**State:**	OH
Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	03/16/2015	**End Date:**	03/15/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	17	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 3/15/2022 per NSSC information (Ed., 3/16/21) NOTE: Extended to 3/15/2021 per NSSC information (Ed., 12/16/19) NOTE: Extended to 3/15/2020 per NSSC information (Ed., 12/17/18) NOTE: Extended to 3/15/2019 per NSSC information (Ed., 12/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AG55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plant growth is characterized by plasticity, allowing environmental cues like light and gravity to play a significant role in determining the overall form of the plant body. Statolith sedimentation in specialized cells represents one way that plants perceive gravity. Plants lacking functional statoliths show reduced but consistent gravity responses, suggesting the presence of a non-statolith sensory system. This proposal seeks to investigate the gravity perception machinery in roots of several genotypes of the higher plant Arabidopsis thaliana by taking advantage of the unique microgravity environment of Earth orbit on the International Space Station (ISS). One thrust of the work described here is to characterize the physiological properties of the putative non-statolith sensory system, including the threshold acceleration force required to activate it, and its capacity to modulate downstream growth regulation machinery. The second thrust is to determine whether the two gravity sensory systems control a single or multiple output systems. Addressing this question requires the microgravity environment in order to quantify the kinetics of gravitropism in roots having or lacking functional statoliths under fractional g conditions. The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions. In particular, the differences in growth responses and gene regulation between wild-type and starchless mutants will help elucidate the contributions of statolith-based sensing to plant gravity responses.
Task Progress:	We completed the extraction of RNA from the selected flight samples in October, 2019. This involved dissecting frozen seedlings into 4 sections: root tip, root base, hypocotyl, and cotyledons, as we did for the ground pilot samples. We collected 4 replicates of each of these tissue types for both genotypes and across the entire range of acceleration treatments, totaling 384 individual RNA samples. The samples were shipped to the Ohio State University - Ohio Agricultural Research and Development Center (OARDC) Molecular and Cellular core facility, where they used low-input cDNA library kits to produce the libraries for sequencing and perform quality control sequencing (shallow read depth). These samples were then shipped to the molecular core facility at Nationwide Children's Hospital (NCH), where they were sequenced on a NovaSeq 6000 instrument with high read depth (40 million reads per sample, paired-end). We anticipated receiving sequence data in April 2020 and processing these data through the bioinformatics pipeline set up for analyzing the pilot project data. Unfortunately, due to the onset of the Covid-19 pandemic, the core facility at NCH shut down for sample sequencing until fall 2020, requiring an additional period of waiting to receive final sequences. We ultimately received the last batch of sequencing in June, 2021. In April and May 2019, 3 undergraduate students identified T-DNA insertion lines available for approximately half of the contrast analysis target list, and they spent the ensuing 3 months testing about 35 lines for gravity phenotypes. These data were presented by undergraduate students at ASGSR 2019 in Denver, CO at the end of November [Ed. Note: 2019 Annual Conference of the American Society for Gravitational and Space Research] and already indicate numerous genes with hitherto unreported gravity phenotypes, greatly expanding our understanding of this response pathway. The pandemic prevented us from continuing this screen in the summer of 2020, but we continued the work in summer 2021. We have started the analysis of the large dataset generated by RNA sequencing (RNA-seq) of the samples grown for 36 h under altered g conditions on the ISS using the European Modular Cultivation System (EMCS) centrifuge. At this time, we have carried out exploratory data analysis in order to begin to understand the most fruitful approaches and comparisons to choose for the identification of differentially expressed genes and pathways. One key observation is the dramatic difference in the overall regulation of gene expression between wild-type (WT) and starchless (pgm-1) mutants. Plotting a heatmap of expression of the 1000 most variable genes reveals this difference in pattern, with WT seedling roots showing much more consistent regulation than starchless mutants. We are currently analyzing all 384 transcriptomes compared amongst each other to infer patterns of plant transcriptional regulation that may be responsive to the gravity cue as a continuum.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Meyers A, Scinto-Madonich N, Wyatt SE, Wolverton C. "Plant Gravitropism: Methods and Protocols." Methods Mol Biol. 2022;2368:233-9. (Plant Gravitropism: Methods and Protocols). https://doi.org/10.1007/978-1-0716-1677-2_15 ; PMID: 34647259 , Jan-2022